

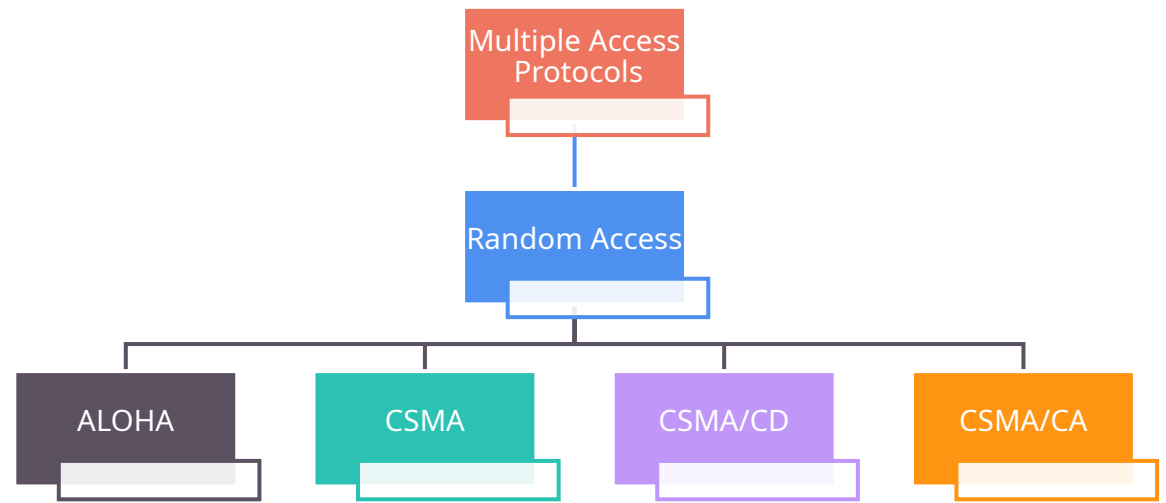
# Access Control

- We can consider the data link layer as two sub layers.
- The upper sub layer is responsible for data link control, and the lower sub layer is responsible for resolving access to the shared media.
- If the channel is dedicated, we do not need the lower sub layer.
- The situation is similar for multipoint networks.
- Many formal protocols have been devised to handle access to a shared little

# Types Of Access Control



# Types Of Access Control

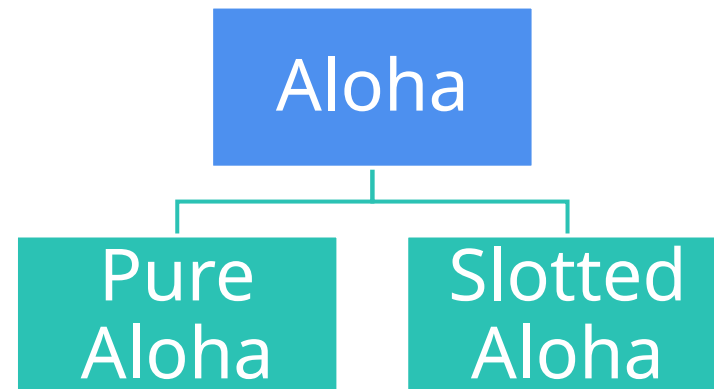


# Random Access Protocol

- In this, all stations have same superiority that is no station has more priority than another station.
- Any station can send data depending on medium's state( idle or busy).
- It has two features:
- There is no fixed time for sending data
- There is no fixed sequence of stations sending data

# Aloha

- It was designed for wireless LAN but is also applicable for shared medium.
- In this, multiple stations can transmit data at the same time and can hence lead to collision and data being garbled.



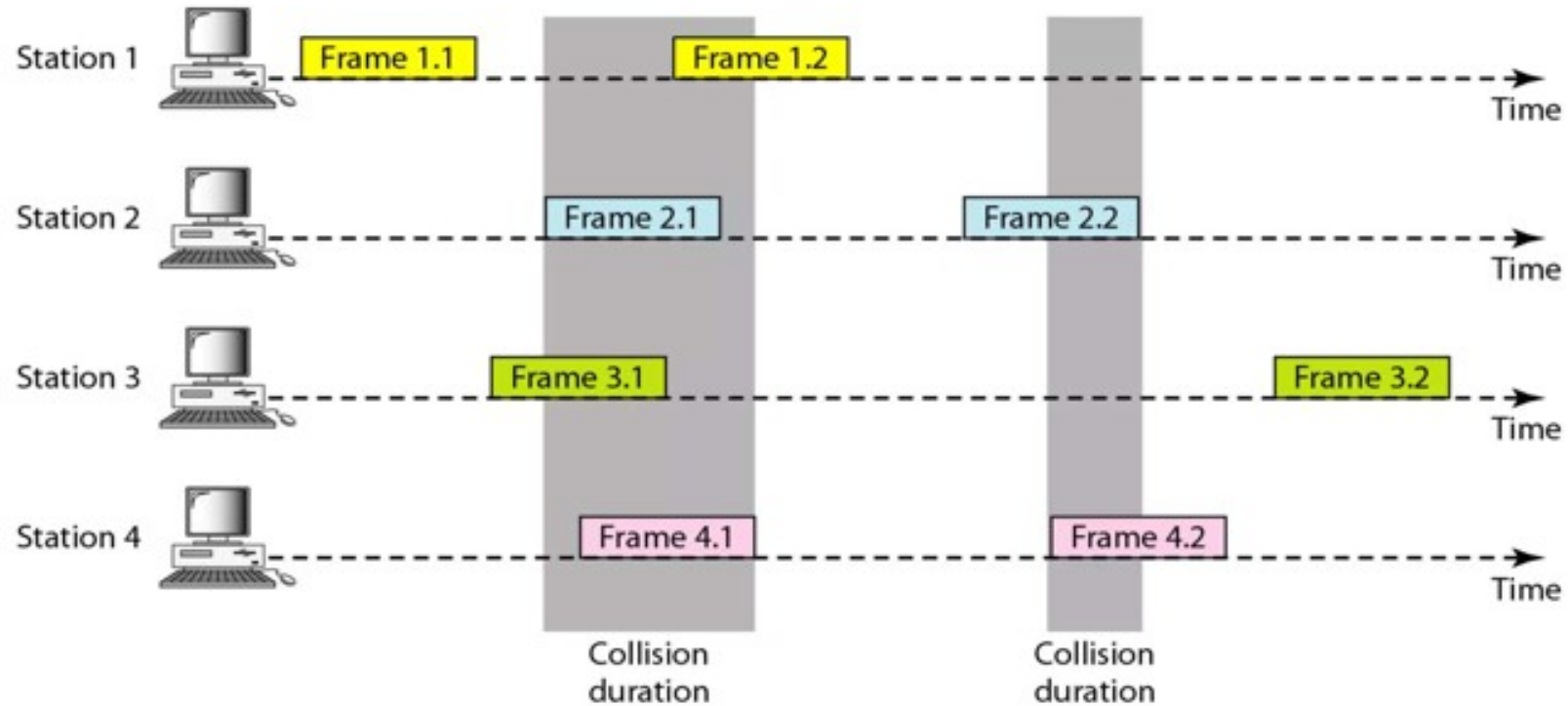
# Pure Aloha

- When a station sends data it waits for an acknowledgement.
- If the acknowledgement doesn't come within the allotted time then the station waits for a random amount of time called back-off time ( $T_b$ ) and re-sends the data.
- Since different stations wait for different amount of time, the probability of further collision decreases.
- It allows the stations to transmit data at any time whenever they want.
- After transmitting the data packet, station waits for some time

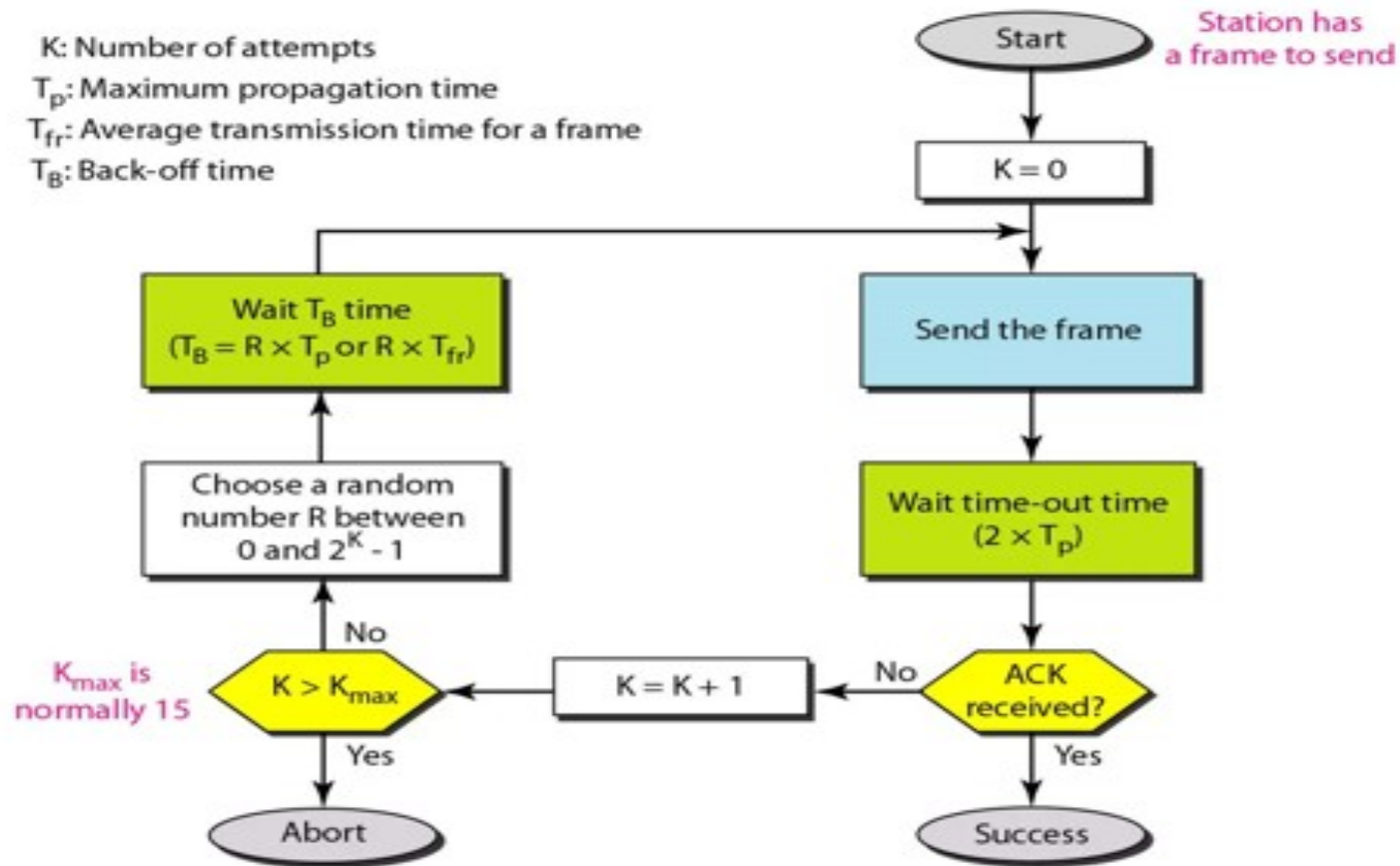
# Pure Aloha

- **Case-01:**
  - Transmitting station receives an acknowledgement from the receiving station.
  - In this case, transmitting station assumes that the transmission is successful.
- **Case-02:**
  - Transmitting station does not receive any acknowledgement within specified time from the receiving station.
  - In this case, transmitting station assumes that the transmission is unsuccessful.

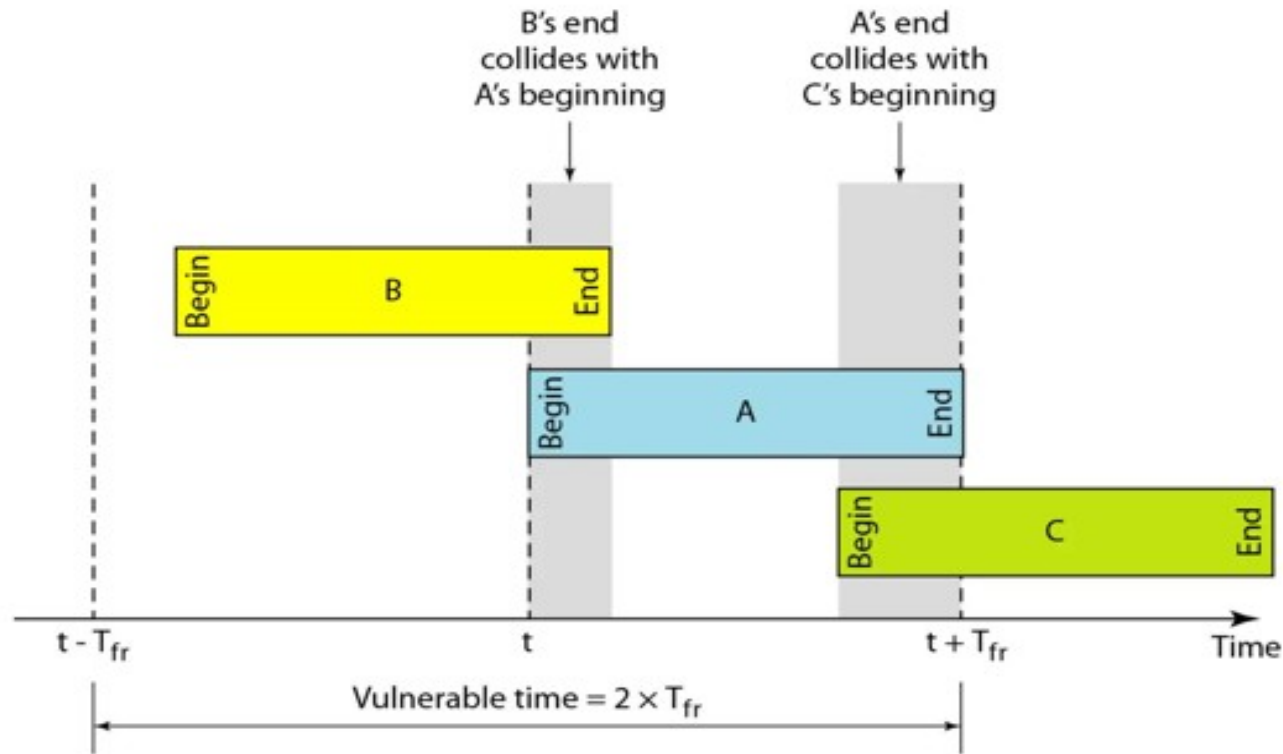
# Frames in Pure Aloha



# Procedure for Pure Aloha



# Vulnerable time for Pure Aloha



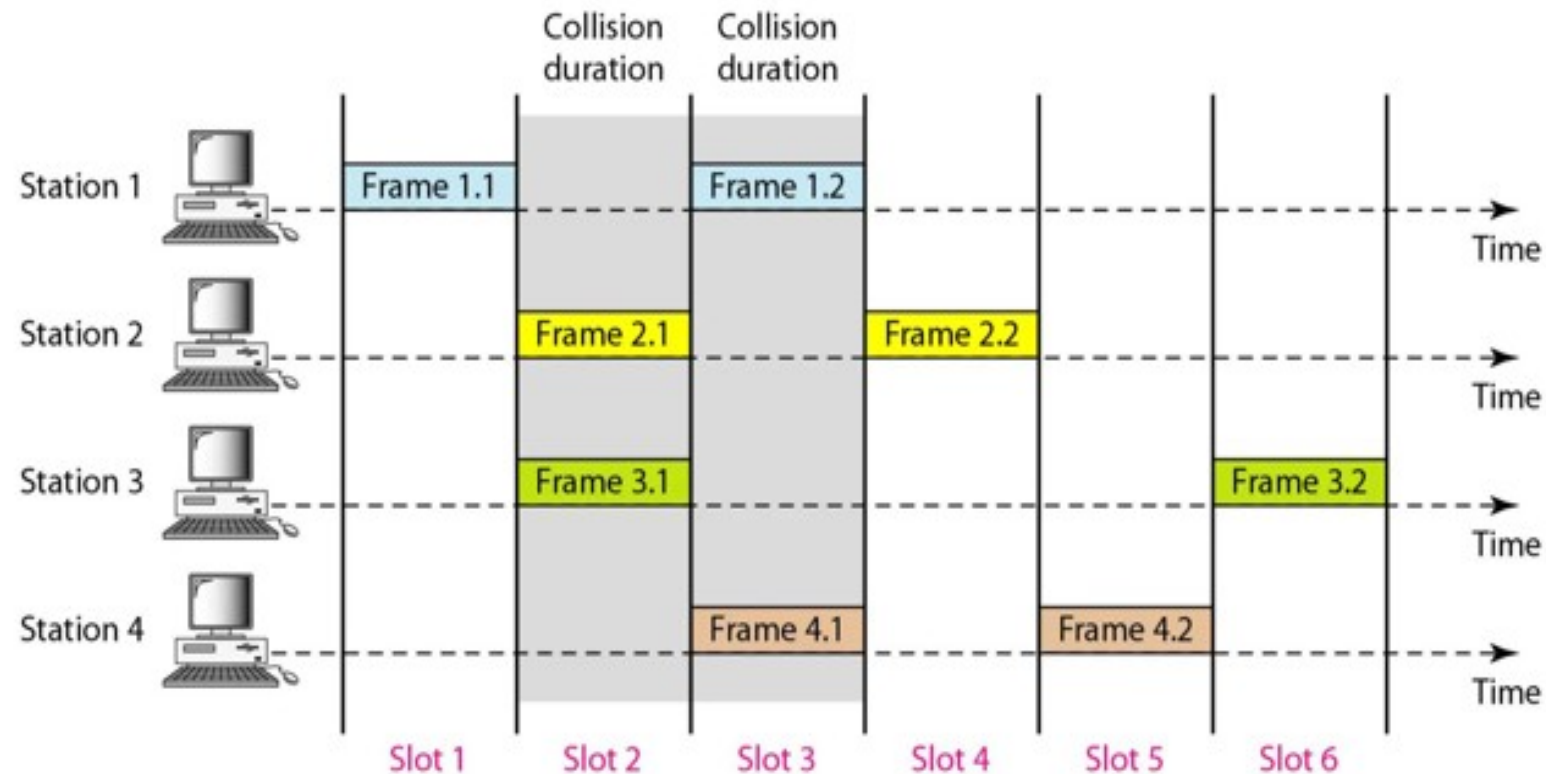
# Pure Aloha

- **Efficiency-**
  - Efficiency of Pure Aloha ( $\eta$ ) =  $G \times e^{-2G}$
  - where  $G$  = Number of stations willing to transmit data.
- **Maximum\_Efficiency-**
  - We put  $d\eta / dG = 0$
  - Maximum value of  $\eta$  occurs at  $G = 1/2$
  - Substituting  $G = 1/2$  in the above expression, we get-
  - Maximum efficiency of Pure Aloha
    - $= 1/2 \times e^{-2 \times 1/2}$
    - $= 1 / 2e$
    - $= 0.184$
    - $= 18.4\%$

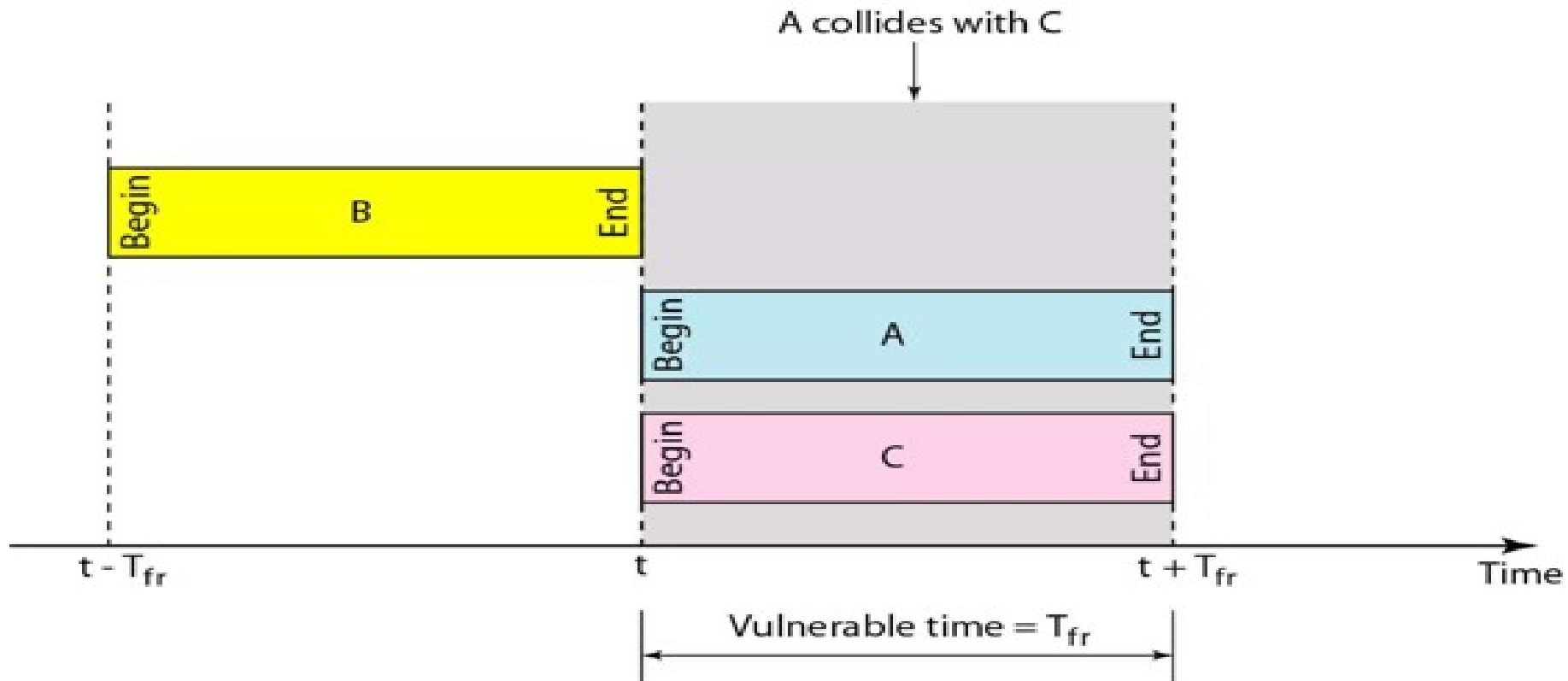
# Slotted Aloha

- Slotted Aloha divides the time of shared channel into discrete intervals called as time slots.
- Any station can transmit its data in any time slot.
- The only condition is that station must start its transmission from the beginning of the time slot.
- If the beginning of the slot is missed, then station has to wait until the beginning of the next time slot.
- A collision may occur if two or more stations try to transmit data at the beginning of the same time slot.

# Frames in Slotted Aloha



# Vulnerable time for Slotted Aloha



# Pure Aloha

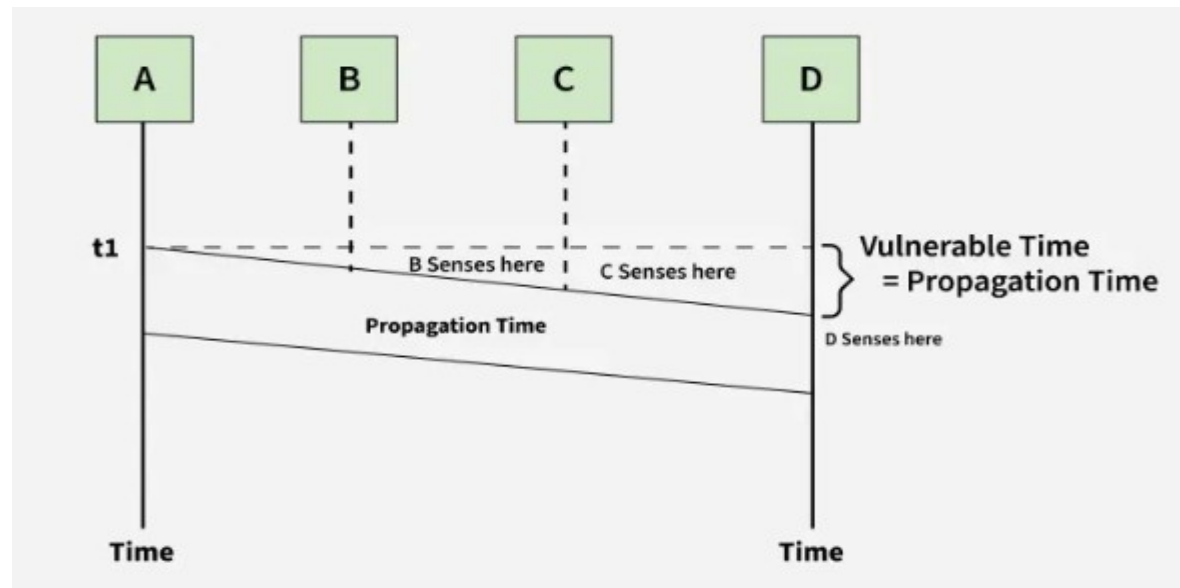
- **Efficiency-**
  - Efficiency of Slotted Aloha ( $\eta$ ) =  $G \times e^{-G}$
  - where  $G$  = Number of stations willing to transmit data.
- **Maximum\_Efficiency-**
  - We put  $d\eta / dG = 0$
  - Maximum value of  $\eta$  occurs at  $G = 1$
  - Substituting  $G = 1$  in the above expression, we get-
    - $= 1 \times e^{-1}$
    - $= 1 / e$
    - $= 0.368$
    - $= 36.8\%$

# CSMA

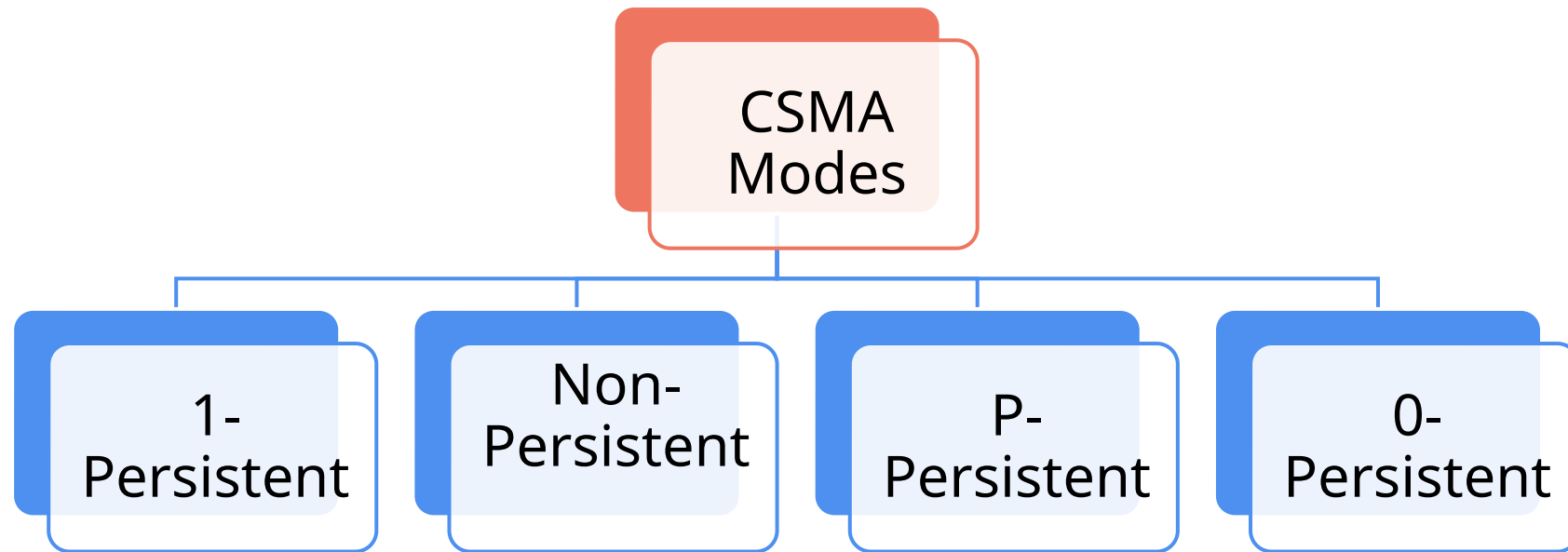
- Carrier Sense Multiple Access ensures fewer collisions as the station is required to first sense the medium (for idle or busy) before transmitting data.
- If it is idle then it sends data, otherwise it waits till the channel becomes idle.
- However there is still chance of collision in CSMA due to propagation delay.
- For example, if station A wants to send data, it will first sense the medium.
- If it finds the channel idle, it will start sending data.

# CSMA

- However, by the time the first bit of data is transmitted (delayed due to propagation delay) from station A, if station B requests to send data and senses the medium it will also find it idle and will also send data.
- This will result in collision of data from station A and B.



# Modes of CSMA



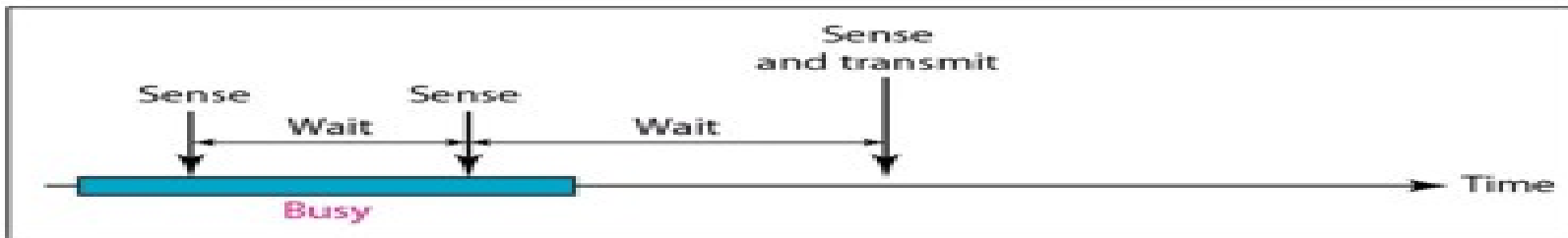
# Modes of CSMA

- **1-Persistent:** The node senses the channel, if idle it sends the data, otherwise it continuously keeps on checking the medium for being idle and transmits unconditionally (with 1 probability) as soon as the channel gets idle.
- **Non-Persistent:** The node senses the channel, if idle it sends the data, otherwise it checks the medium after a random amount of time (not continuously) and transmits when found idle.
- **P-Persistent:** The node senses the medium, if idle it sends the data with  $p$  probability. If the data is not transmitted ( $(1-p)$  probability) then it waits for some time and checks the medium again, now if it is found idle then it send with  $p$  probability. This repeat continues until the frame is sent. It is used in Wifi and packet radio systems.
- **O-Persistent:** Superiority of nodes is decided beforehand and transmission occurs in that order. If the medium is idle, node waits for its time slot to send data.

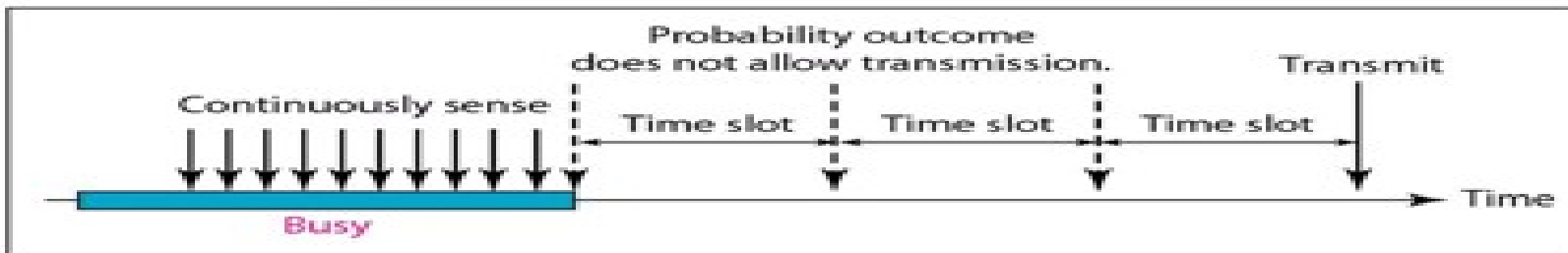
# Behavior of three persistence methods



a. 1-persistent

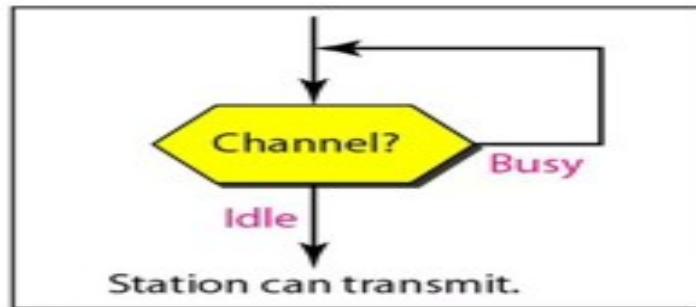


b. Nonpersistent

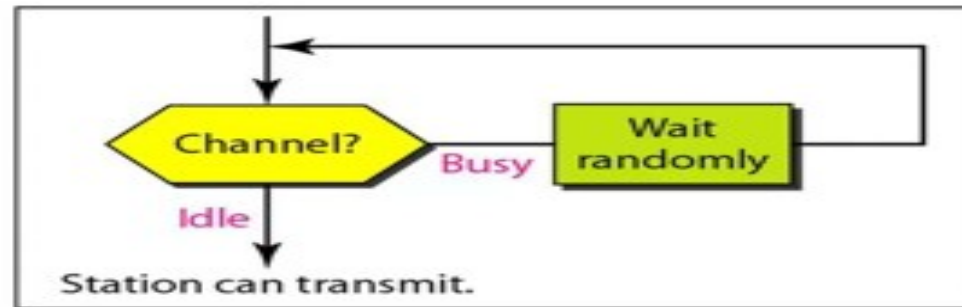


c. p-persistent

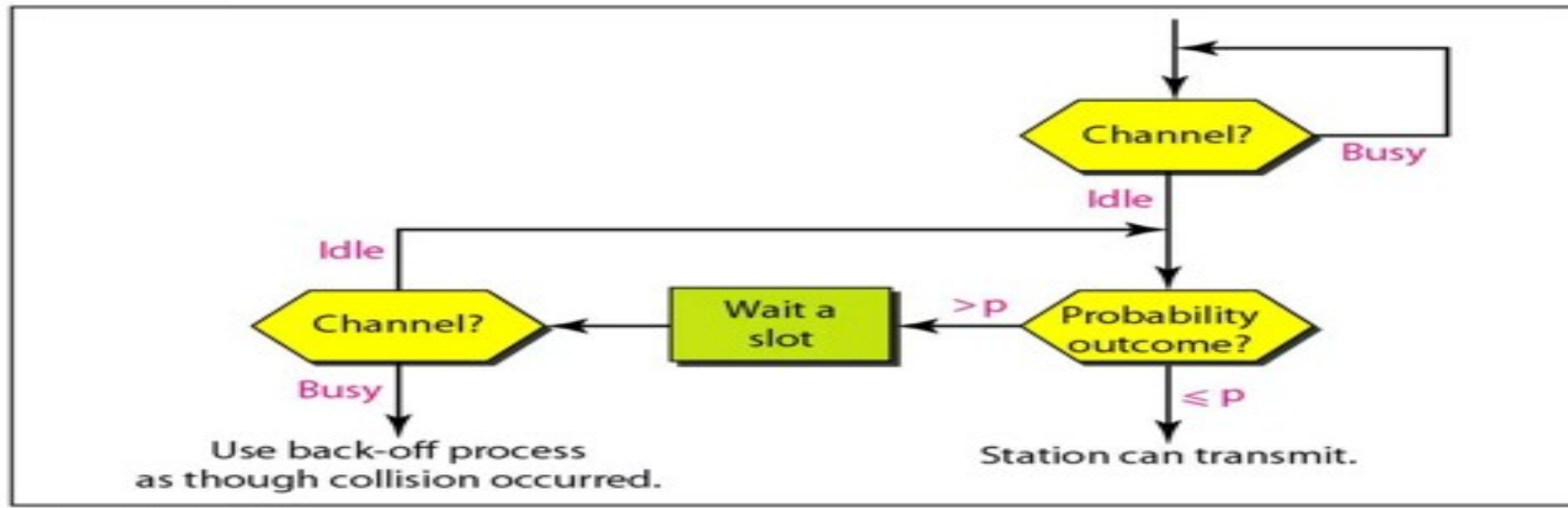
# Flow Diag. of three persistence methods



a. 1-persistent

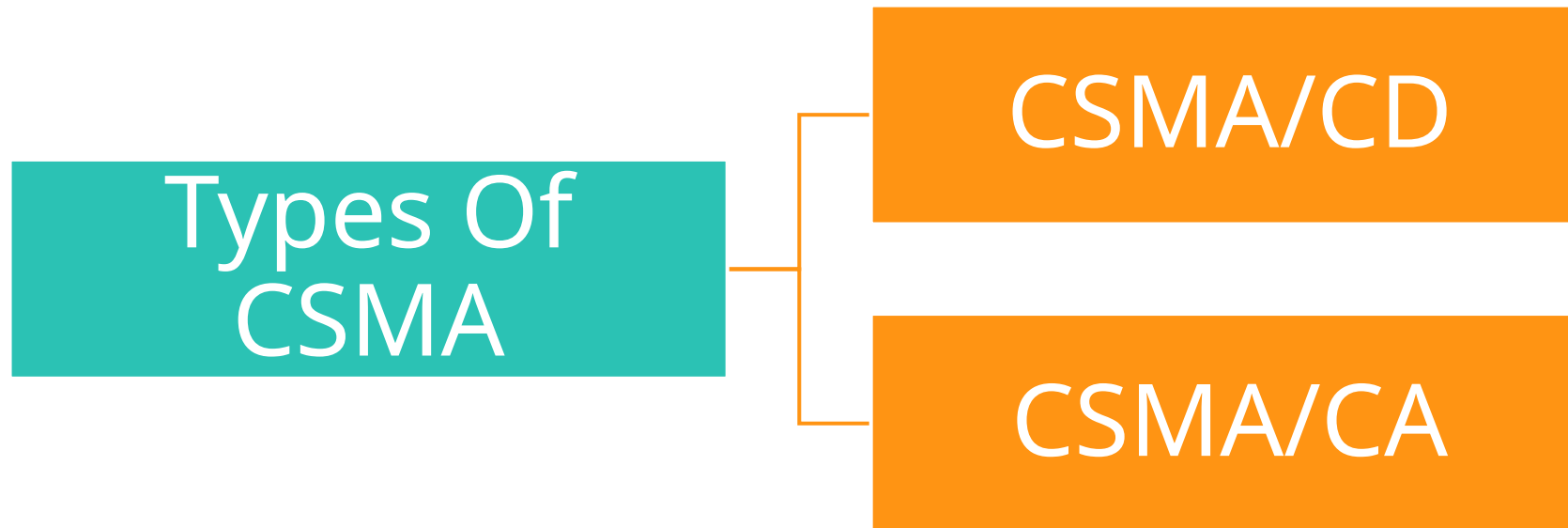


b. Nonpersistent



c. p-persistent

# Types of CSMA

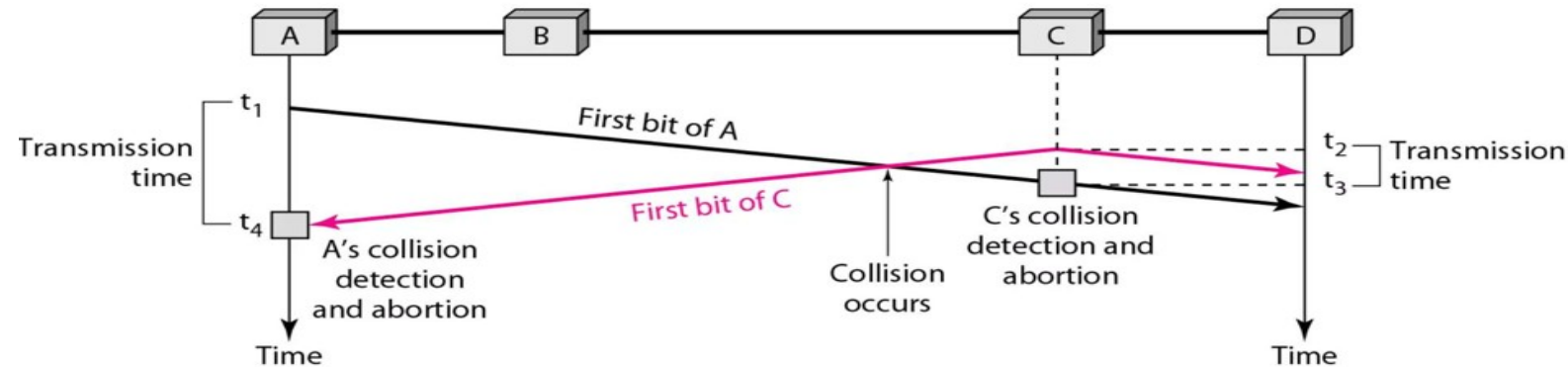


# CSMA/CD

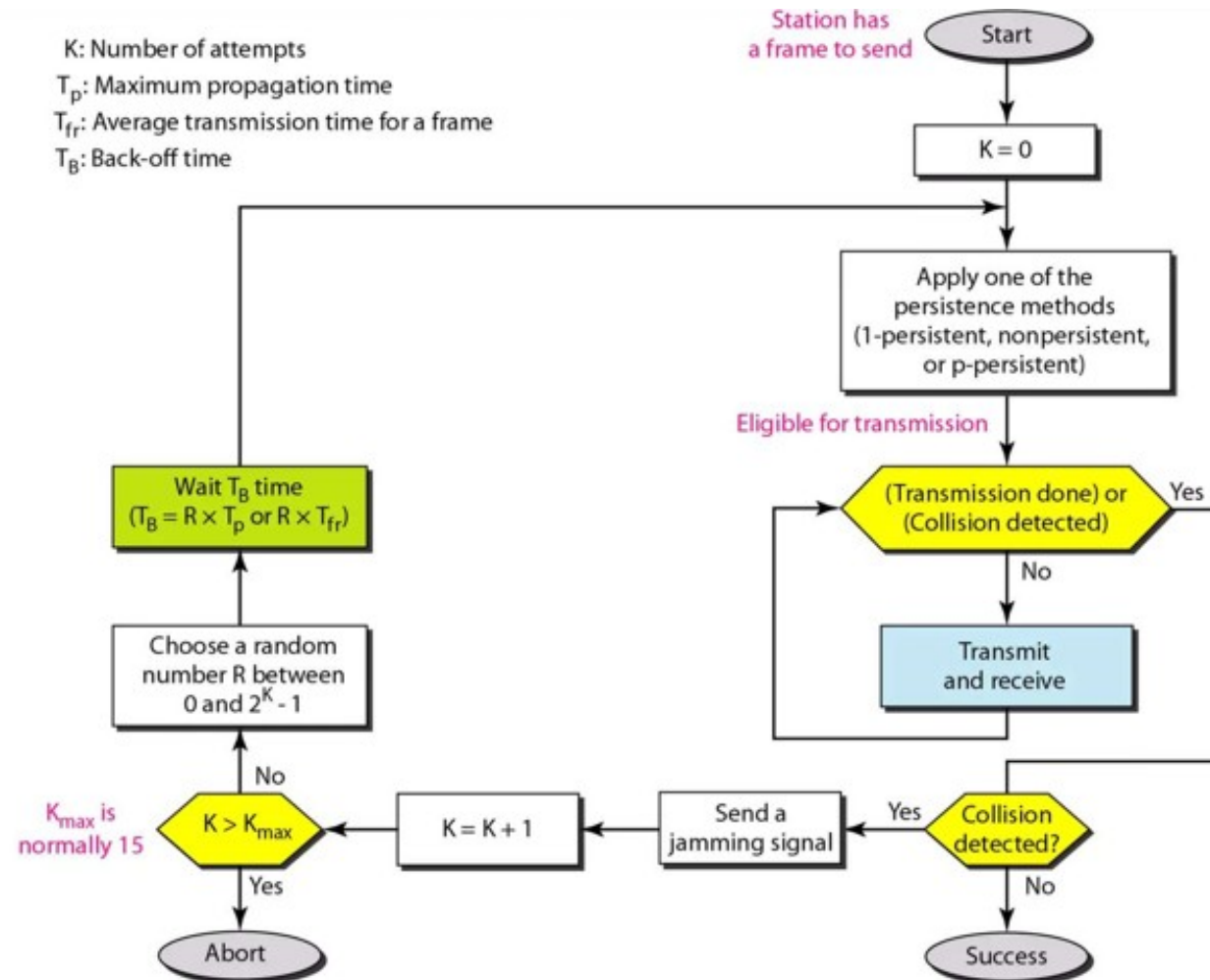
- CSMA / CD stands for Carrier Sense Multiple Access / Collision Detection.
- In this method, a station monitors the medium after it sends a frame to see if the transmission was successful.
- If so, the station is finished.
- If, however, there is a collision, the frame is sent again.
- Each station can sense the carrier only at its point of contact with the carrier.
- It is not possible for any station to sense the entire carrier.
- Thus, there is a huge possibility that a station might sense the carrier free even when it is actually not.

# CSMA/CD

- To better understand CSMA/CD, let us look at the first bits transmitted by the two stations involved in the collision.
- Although each station continues to send bits in the frame until it detects the collision, we show what happens as the first bits collide.



# Procedure for CSMA/CD



# CSMA/CD

## Case-01:

- If no collided signal comes back during the transmission,
- It indicates that no collision has occurred.
- The data packet is transmitted successfully.

## Case-02:

- If the collided signal comes back during the transmission,
- It indicates that the collision has occurred.
- The data packet is not transmitted successfully.
- Step-03 is followed.

# CSMA/CD

## Releasing Jam Signal

- Jam signal is a 48 bit signal.
- It is released by the transmitting stations as soon as they detect a collision.
- It alerts the other stations not to transmit their data immediately after the collision.
- Otherwise, there is a possibility of collision again with the same data packet.
- Ethernet sends the jam signal at a frequency other than the frequency of data signals.
- This ensures that jam signal does not collide with the data signals undergone collision.

# CSMA/CD

## Waiting For Back Off Time

- After the collision, the transmitting station waits for some random amount of time called as back off time.
- After back off time, it tries transmitting the data packet again.
- If again the collision occurs, then station again waits for some random back off time and then tries again.
- The station keeps trying until the back off time reaches its limit.
- After the limit is reached, station aborts the transmission.

# CSMA/CD

## Note-01

- CSMA / CD is used in wired LANs.
- CSMA / CD is standardized in IEEE 802.3

## Note-02

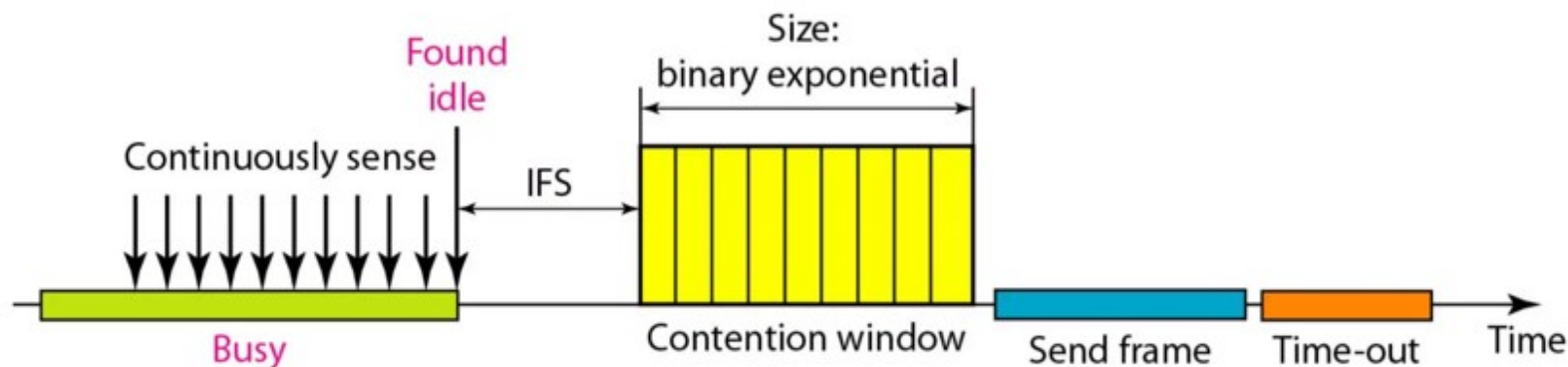
- CSMA / CD only minimizes the recovery time.
- It does not take any steps to prevent the collision until it has taken place.

# CSMA/CA

- CSMA/CA is a technique used mainly in wireless networks to avoid collisions when multiple devices try to send data.
- Unlike wired networks where a device can detect collisions easily, in wireless networks, most of the energy goes into transmitting, so a device cannot sense if a collision has happened during transmission.
- To solve this, CSMA/CA prevents collisions instead of detecting them, making it ideal for technologies like Wi-Fi.

# CSMA/CA

- Carrier sense multiple access with collision avoidance was invented for this network.
- Collisions are avoided through the use of CSMA/CA's three strategies: the inter-frame space, the contention window, and acknowledgments, as shown in Figure.



# Strategies used in CSMA/CA

- **Inter Frame Space (IFS)**

- After the channel becomes idle, the station waits for a short fixed time to avoid sudden collisions shorter IFS means higher priority.

- **Contention Window:**

The station picks a random slot within a time window to wait before transmitting, helping spread out transmissions and reduce collision chances.

- **Acknowledgments (ACKs):**

The receiver sends back an ACK if data is received correctly; if not received in time, the sender assumes failure and tries again.

# Procedure for CSMA/CA



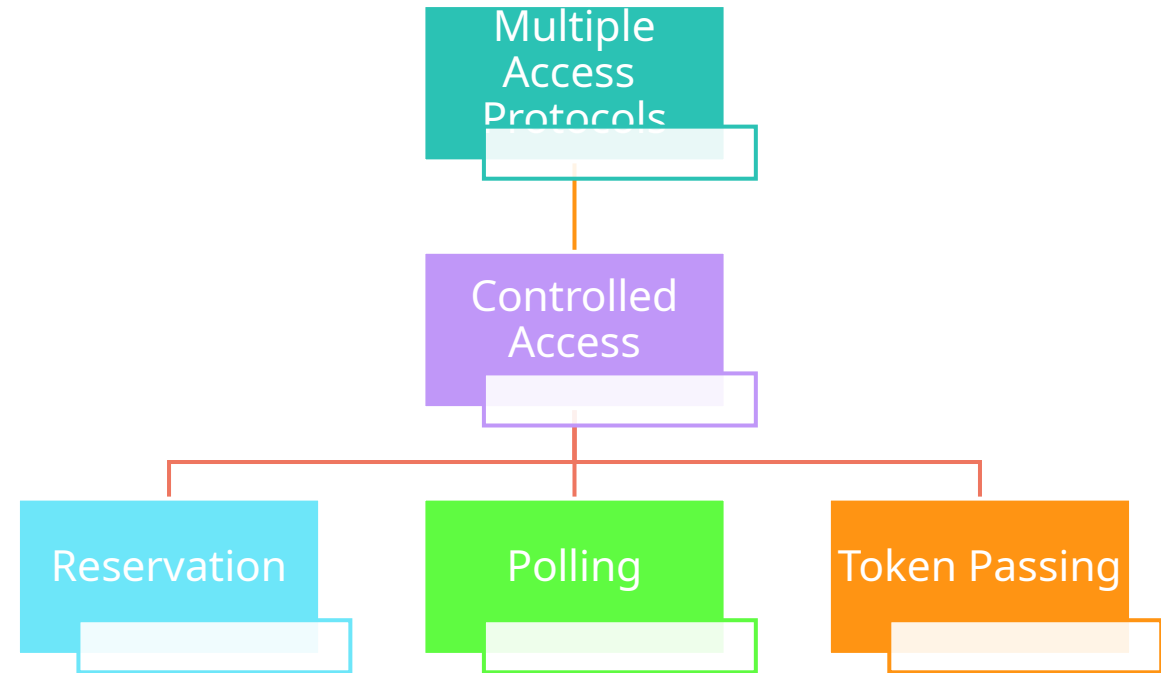
# CSMA/CA

- **Initialize Attempt Counter:** Set the number of transmission attempts,  $K = 0$ .
- **Check if Channel is Idle:** Listen to the channel; if it's busy, keep checking until it's idle.
- **Wait Inter-frame Space (IFS) Time:** After the channel is idle, wait for the IFS period to ensure fairness.
- **Check Idle Again After IFS:** If the channel is still idle after IFS, proceed; otherwise, restart from Step 2.
- **Generate Random Back-off:** Choose a random number  $R$  between 0 and  $2K - 1$  for back-off slots.

# CSMA/CA

- **Wait for R Time Slots:** Defer transmission by waiting for R time slots to avoid collisions.
- **Transmit Frame:** After the backoff, send the data frame.
- **Wait for Acknowledgment (ACK):** Wait to see if an ACK is received from the receiver.
- **ACK Received:** If Yes then Transmission was successful; end process , or if No then Go to Step 10.
- **Check Retry Limit:**  
If  $K > K_{max}$ , abort transmission (too many attempts).  
If  $K \leq K_{max}$ , increment K by 1 and go back to Step 2.

# Types Of Access Control



# CONTROLLED ACCESS

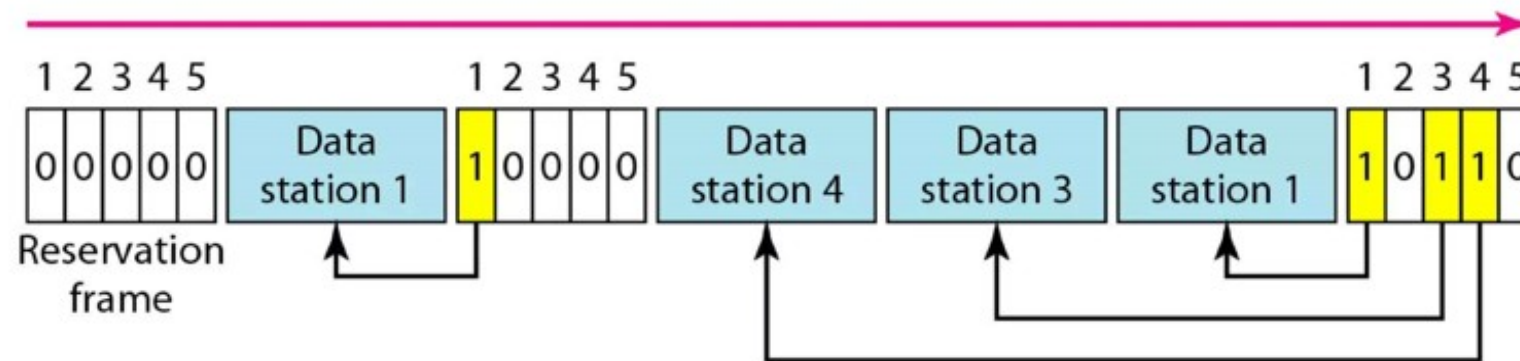
- In controlled access, the stations consult one another to find which station has the right to send.
- A station cannot send unless it has been authorized by other stations.
- We discuss three popular controlled-access methods.
- Reservation
- Polling
- Token Passing

# Reservation

- In the reservation method, a station needs to make a reservation before sending data.
- Time is divided into intervals.
- In each interval, a reservation frame precedes the data frames sent in that interval.
- If there are  $N$  stations in the system, there are exactly  $N$  reservation mini-slots in the reservation frame.
- Each mini-slot belongs to a station.
- When a station needs to send a data frame, it makes a reservation in its own mini-slot.

# Reservation

- The stations that have made reservations can send their data frames after the reservation frame.
- Figure shows a situation with five stations and a five mini-slot reservation frame.
- In the first interval, only stations 1, 3, and 4 have made reservations.
- In the second interval, only station 1 has made a reservation.

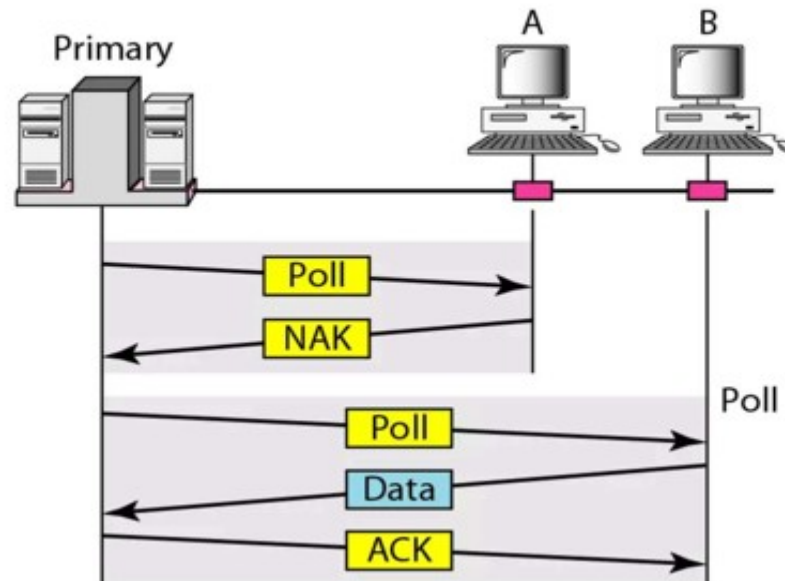
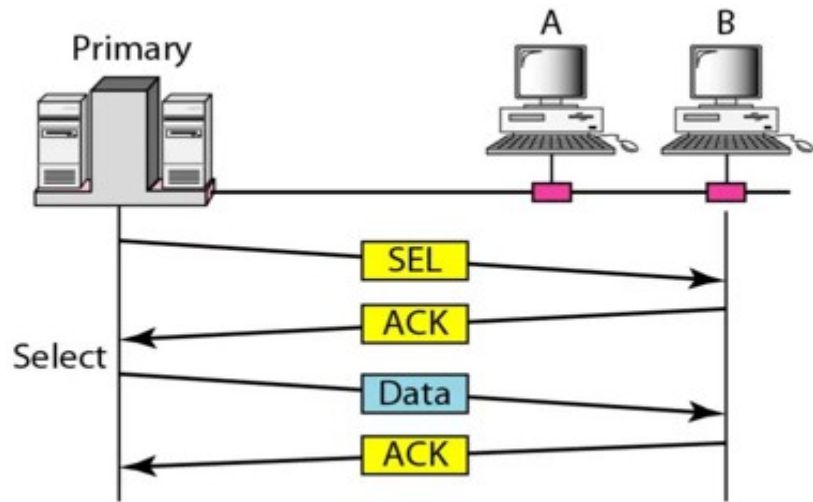


# Polling

- A polling is conducted in which all the stations willing to send data participates.
- The polling algorithm chooses one of the stations to send the data.
- The chosen station sends the data to the destination.
- After the chosen station has sent the data, the cycle repeats.
- Polling works with topologies in which one device is designated as a primary station and the other devices are secondary stations.
- All data exchanges must be made through the primary device even when the ultimate destination is a secondary device.
- The primary device controls the link; the secondary devices follow its instructions.

# Polling

- It is up to the primary device to determine which device is allowed to use the channel at a given time.
- The primary device, therefore, is always the initiator of a session (see Figure select and poll function).



# Select Function

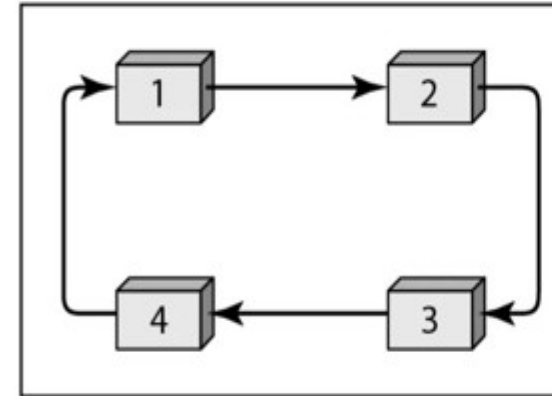
- The select function is used whenever the primary device has something to send.
- Remember that the primary controls the link.
- If the primary is neither sending nor receiving data, it knows the link is available.
- If it has something to send, the primary device sends it.
- What it does not know, however, is whether the target device is prepared to receive.
- So the primary must alert the secondary to the upcoming transmission and wait for an acknowledgment of the secondary's ready status.
- Before sending data, the primary creates and transmits a select (SEL) frame, one field of which includes the address of the intended secondary.

# Poll Function

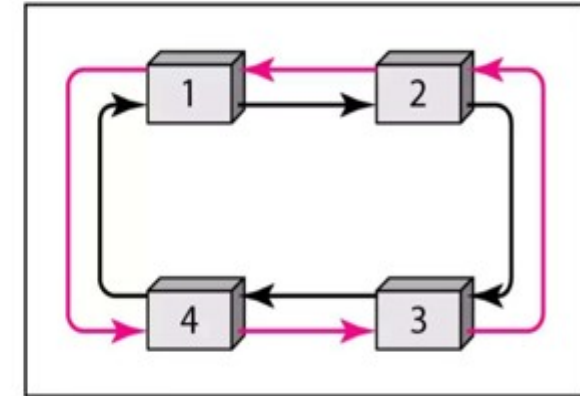
- The poll function is used by the primary device to solicit transmissions from the secondary devices.
- When the primary is ready to receive data, it must ask (poll) each device in turn if it has anything to send.
- When the first secondary is approached, it responds either with a NAK frame if it has nothing to send or with data (in the form of a data frame) if it does.
- If the response is negative (a NAK frame), then the primary polls the next secondary in the same manner until it finds one with data to send.
- When the response is positive (a data frame), the primary reads the frame and returns an acknowledgment (ACK frame), verifying its receipt.

# Token Passing

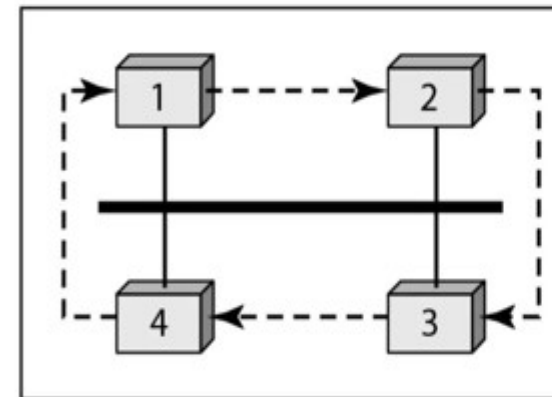
- All the stations are logically connected to each other in the form of a ring.
- The access of stations to the transmission link is governed by a token.
- A station is allowed to transmit a data packet if and only if it possess the token otherwise not.
- Each station passes the token to its neighboring station either clockwise or anti-clockwise.



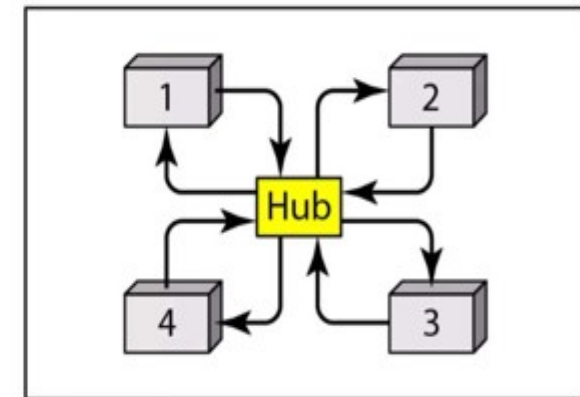
a. Physical ring



b. Dual ring



c. Bus ring



d. Star ring

# Token Passing

- In the token-passing method, the stations in a network are organized in a logical ring.
- In other words, for each station, there is a predecessor and a successor.
- The predecessor is the station which is logically before the station in the ring; the successor is the station which is after the station in the ring.
- The current station is the one that is accessing the channel now.
- The right to this access has been passed from the predecessor to the current station.
- The right will be passed to the successor when the current station has no more data to send.

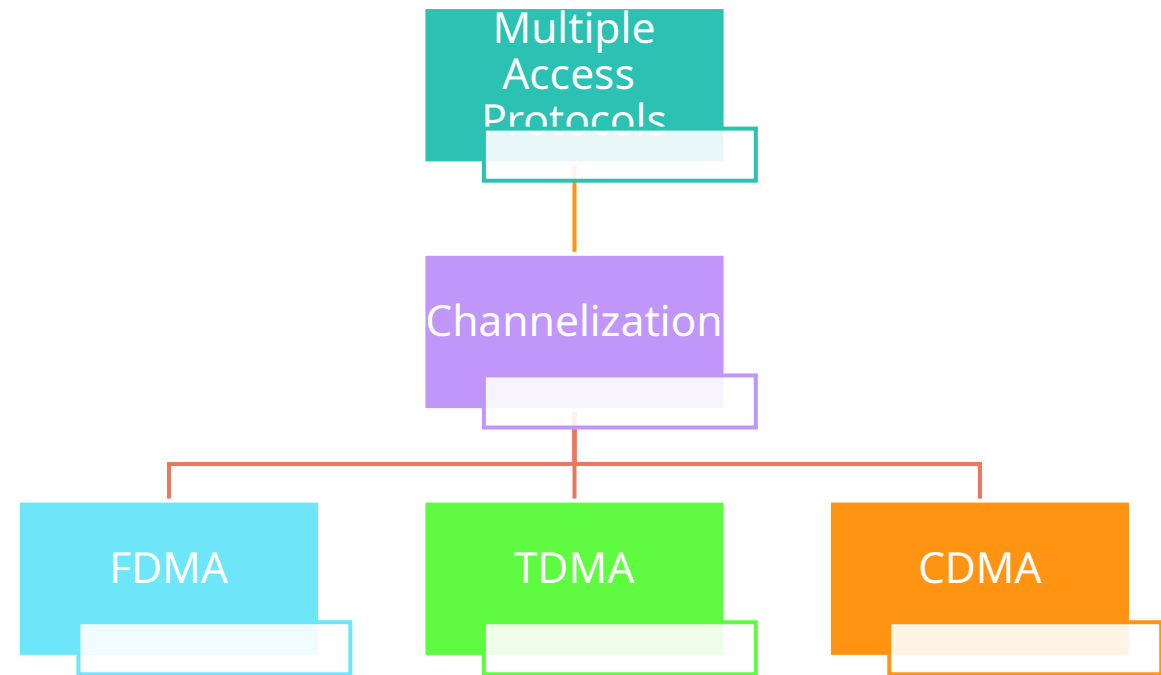
# Token Passing

- In this method, a special packet called a token circulates through the ring.
- The possession of the token gives the station the right to access the channel and send its data.
- When a station has some data to send, it waits until it receives the token from its predecessor.
- It then holds the token and sends its data.
- When the station has no more data to send, it releases the token, passing it to the next logical station in the ring.
- The station cannot send data until it receives the token again in the next round.

# Token Passing

- In this process, when a station receives the token and has no data to send, it just passes the data to the next station.
- Stations must be limited in the time they can have possession of the token.
- The token must be monitored to ensure it has not been lost or destroyed.
- For example, if a station that is holding the token fails, the token will disappear from the network.
- Another function of token management is to assign priorities to the stations and to the types of data being transmitted.
- And finally, token management is needed to make low-priority stations release the token to high priority stations.

# Types Of Access Control



# CHANNELIZATION

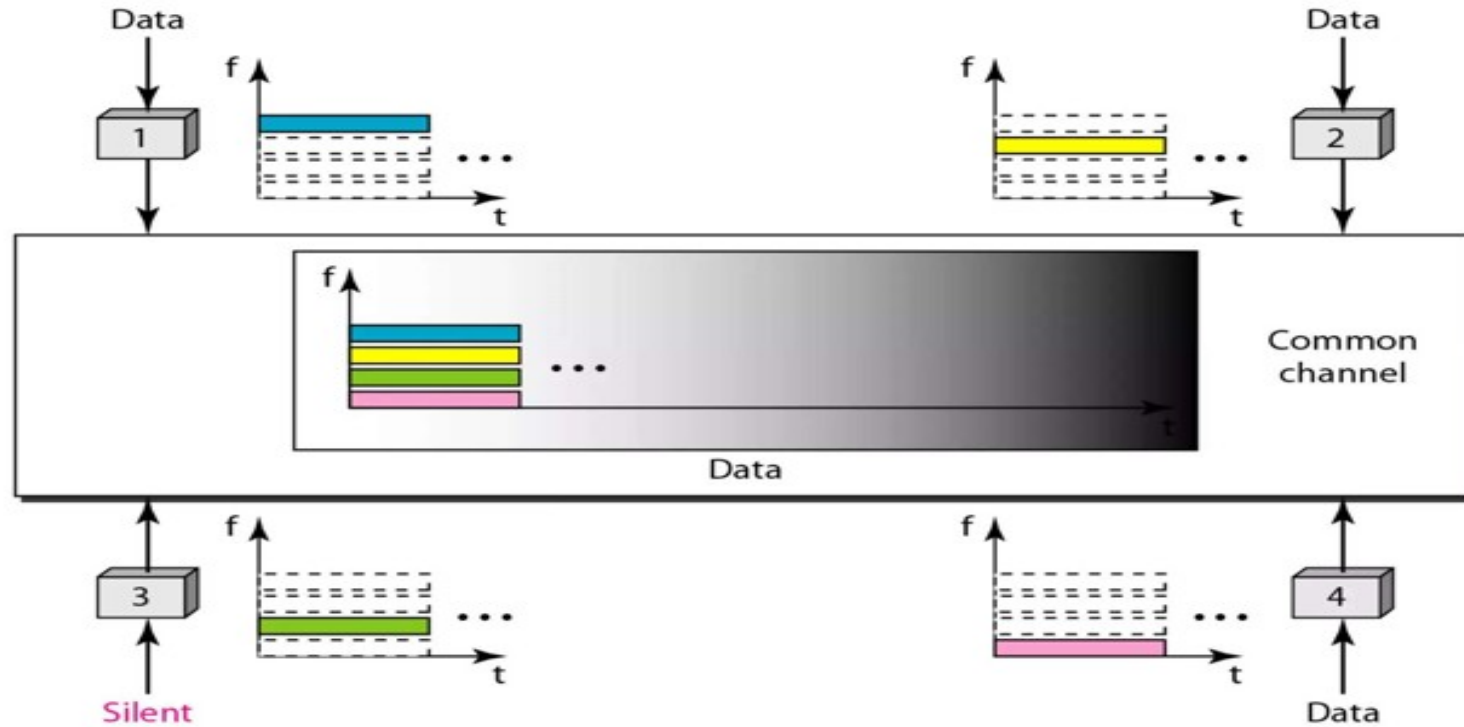
- Channelization is a multiple-access method in which the available bandwidth of a link is shared in time, frequency, or through code, between different stations.
- In this section, we discuss three channelization protocols:
  - FDMA
  - TDMA
  - CDMA

# FDMA

- In frequency-division multiple access (FDMA), the available bandwidth is divided into frequency bands.
- Each station is allocated a band to send its data.
- In other words, each band is reserved for a specific station, and it belongs to the station all the time.
- Each station also uses a band pass filter to confine the transmitter frequencies.
- To prevent station interferences, the allocated bands are separated from one another by small guard bands.

# FDMA

- In FDMA, the available bandwidth of the common channel is divided into bands that are separated by guard bands.

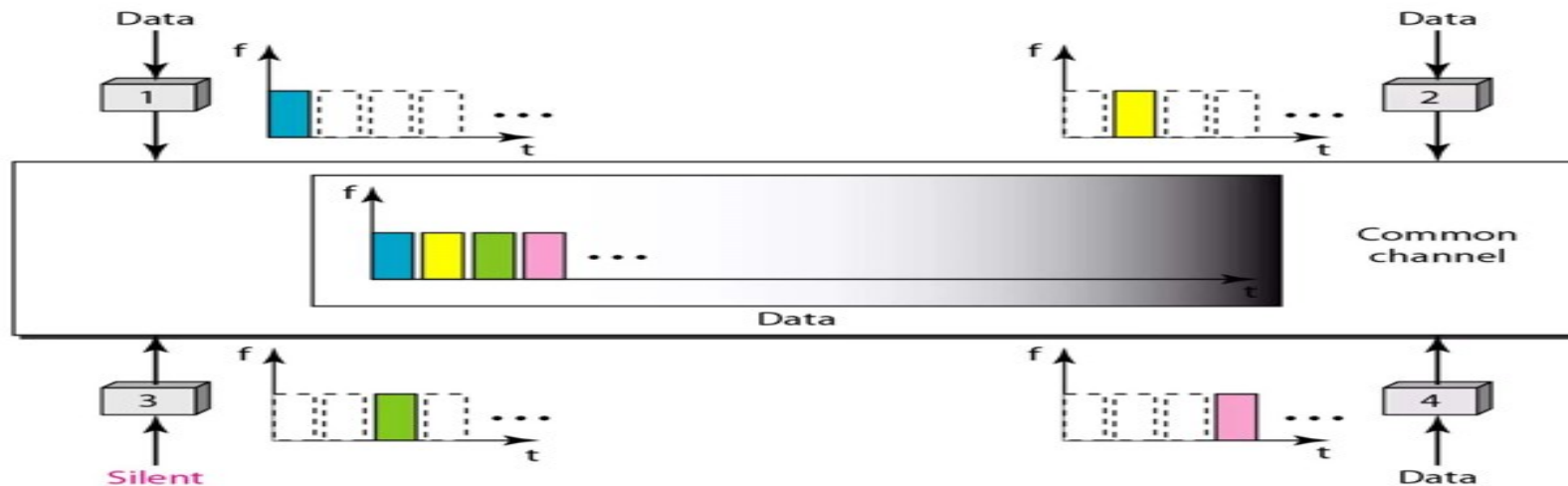


# FDMA

- FDMA specifies a predetermined frequency band for the entire period of communication.
- This means that stream data (a continuous flow of data that may not be packetized) can easily be used with FDMA.
- The data link layer in each station tells its physical layer to make a band-pass signal from the data passed to it.
- The signal must be created in the allocated band. The signals created at each station are automatically band-pass filtered.
- They are mixed when they are sent to the common channel.

# TDMA

- In time-division multiple access (TDMA), the stations share the bandwidth of the channel in time.
- Each station is allocated a time slot during which it can send data.
- Each station transmits its data in its assigned time slot. Figure shows the idea behind TDMA.



# TDMA

- In TDMA, the bandwidth is just one channel that is timeshared between different stations.
- The main problem with TDMA lies in achieving synchronization between the different stations.
- Each station needs to know the beginning of its slot and the location of its slot.
- This may be difficult because of propagation delays introduced in the system if the stations are spread over a large area.
- To compensate for the delays, we can insert guard times. Synchronization is normally accomplished by having some synchronization bits (normally refined to as preamble bits) at the beginning of each slot.

# TDMA

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# CDMA

- In CDMA, one channel carries all transmissions simultaneously.
- Code-division multiple access (CDMA) was conceived several decades ago.
- Recent advances in electronic technology have finally made its implementation possible.
- CDMA differs from FDMA because only one channel occupies the entire bandwidth of the link.
- It differs from TDMA because all stations can send data simultaneously; there is no timesharing.

# CDMA

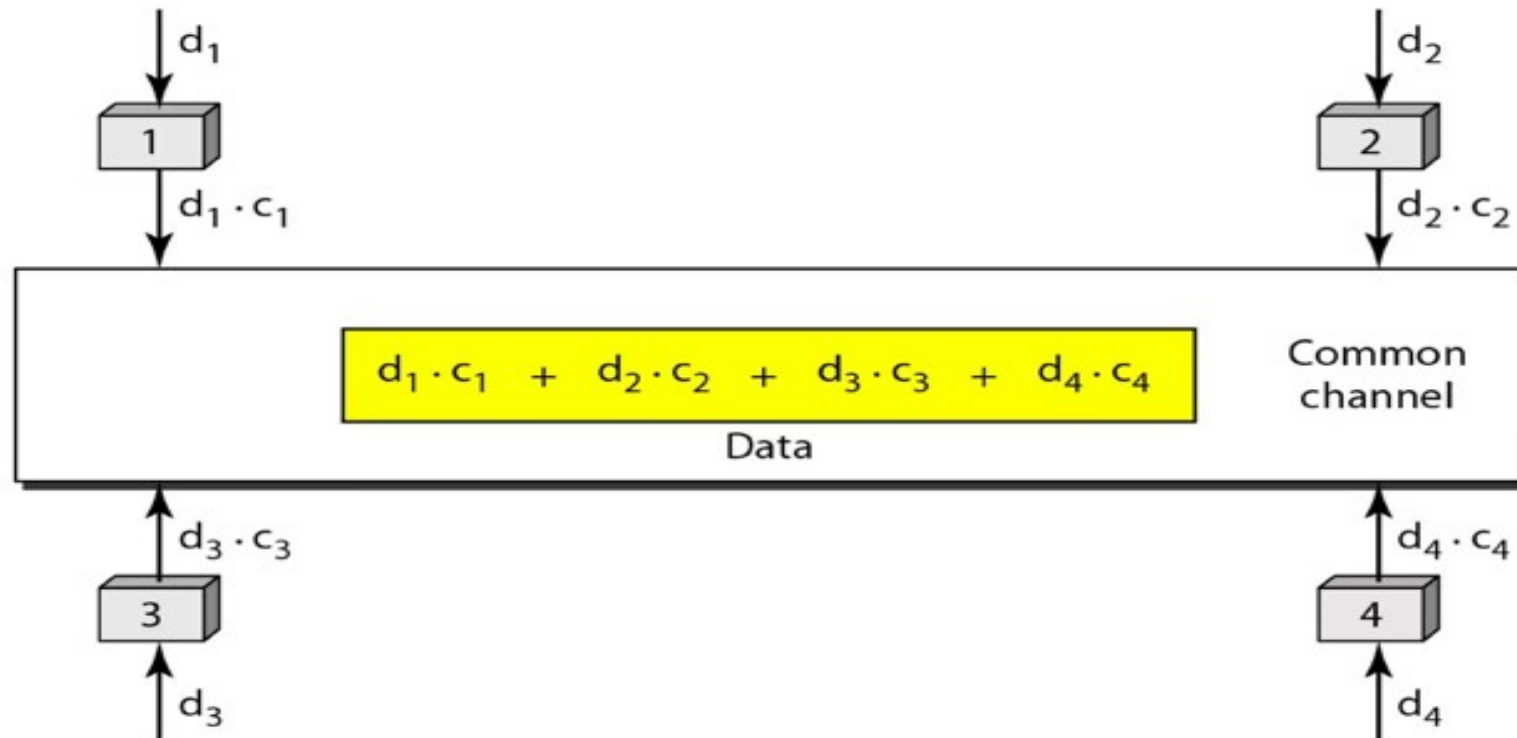
- **Analogy**
- CDMA simply means communication with different codes.
- For example, in a large room with many people, two people can talk in English if nobody else understands English.
- Another two people can talk in Chinese if they are the only ones who understand Chinese, and so on.
- In other words, the common channel, the space of the room in this case, can easily allow communication between several couples, but in different languages (codes).

# CDMA

- **Idea**
- Let us assume we have four stations 1, 2, 3, and 4 connected to the same channel.
- The data from station 1 are  $d_1$ , from station 2 are  $d_2$ , and so on.
- The code assigned to the first station is  $c_1$ , to the second is  $c_2$ , and so on.
- We assume that the assigned codes have two properties.
  1. If we multiply each code by another, we get 0.
  2. If we multiply each code by itself, we get 1 (the number of stations).

# CDMA

- With these two properties in mind, let us see how the above four stations can send data using the same common channel, as shown in Figure.



# CDMA

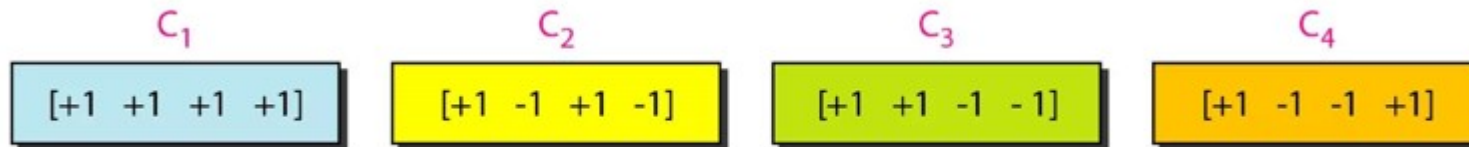
- Station 1 multiplies (a special kind of multiplication, as we will see) its data by its code to get  $d_1$ .
- Station 2 multiplies its data by its code to get  $d_2$ . And so on.
- The data that go on the channel are the sum of all these terms, as shown in the box.
- Any station that wants to receive data from one of the other three multiplies the data on the channel by the code of the sender.

# CDMA

- For example, suppose stations 1 and 2 are talking to each other.
- Station 2 wants to hear what station 1 is saying. It multiplies the data on the channel by  $c_1$  the code of station 1.
- Because  $(c_1 \cdot c_1)$  is 4, but  $(c_2 \cdot c_1)$ ,  $(c_3 \cdot c_1)$ , and  $(c_4 \cdot c_1)$  are all 0s, station 2 divides the result by 4 to get the data from station 1.
- $\text{data} = (d_1 \cdot C_1 + d_2 \cdot C_2 + d_3 \cdot C_3 + d_4 \cdot C_4) \cdot C_1$
- $= d_1 \cdot C_1 \cdot C_1 + d_2 \cdot C_2 \cdot C_1 + d_3 \cdot C_3 \cdot C_1 + d_4 \cdot C_4 \cdot C_1$
- $= 4 \times d_1$

# CDMA

- **Chips**
- CDMA is based on coding theory. Each station is assigned a code, which is a sequence of numbers called chips, as shown in Figure .
- The codes are for the previous example.



# CDMA

They are called orthogonal sequences and have the following properties:

1. Each sequence is made of N elements, where N is the number of stations.
2. If we multiply a sequence by a number, every element in the sequence is multiplied by that element.
  - This is called multiplication of a sequence by a scalar.
  - For example,
  - 2.  $[+1 +1 -1 -1] = [+2 +2 -2 -2]$

# CDMA

3. If we multiply two equal sequences, element by element, and add the results, we get N, where N is the number of elements in the each sequence.

- This is called the inner product of two equal sequences.
- For example,
- $[+1 +1 -1 -1] \cdot [+1 +1 -1 -1] = 1 + 1 + 1 + 1 = 4$

4. If we multiply two different sequences, element by element, and add the results, we get 0.

- This is called inner product of two different sequences.

# CDMA

- For example,
- $[+1 \ +1 \ -1 \ -1] \cdot [+1 \ +1 \ +1 \ +1] = 1 + 1 - 1 - 1 = 0$

5. Adding two sequences means adding the corresponding elements.

The result is another sequence.

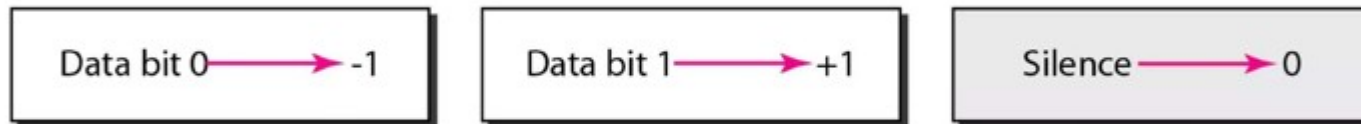
For example,

- $[+1+1-1-1]+[+1+1+1+1]=[+2+2 \ 00]$

# CDMA

## Data Representation

- We follow these rules for encoding:
- If a station needs to send a 0 bit, it encodes it as -1;
- if it needs to send a 1 bit, it encodes it as +1.
- When a station is idle, it sends no signal, which is interpreted as a 0.
- These are shown in Figure1



# CDMA

## Encoding and Decoding

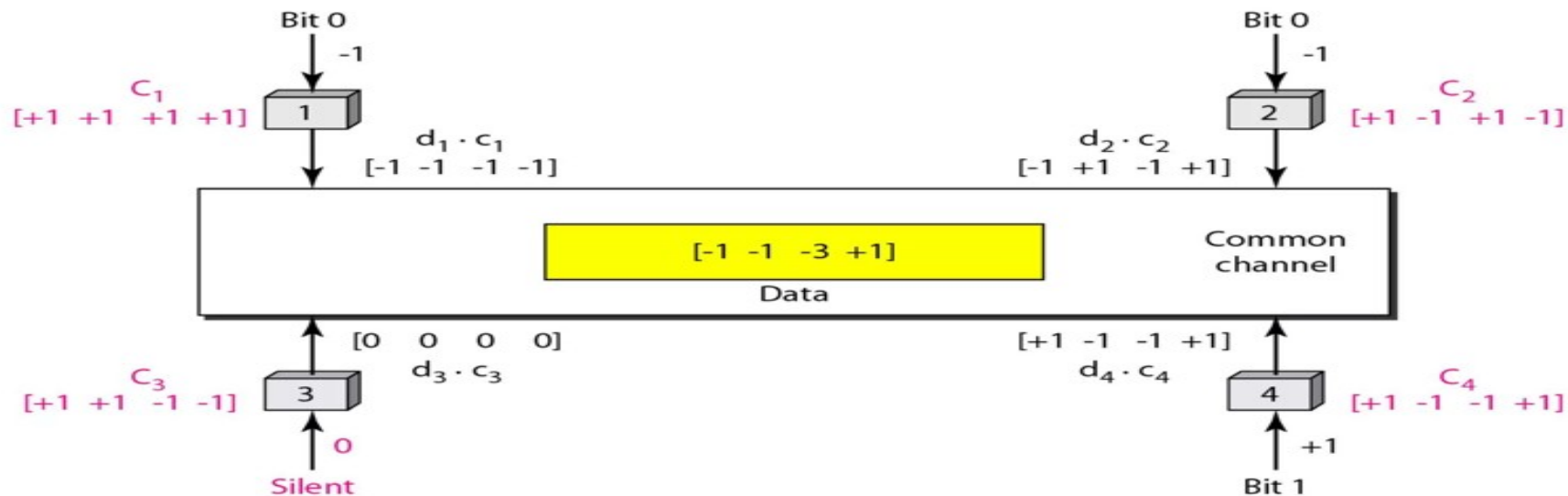
- As a simple example, we show how four stations share the link during a 1-bit interval.
- The procedure can easily be repeated for additional intervals.
- We assume that stations 1 and 2 are sending a 0 bit and channel 4 is sending a 1 bit.
- Station 3 is silent.
- The data at the sender site are translated to -1, -1, 0, and +1.
- result is a new sequence which is sent to the channel.

# CDMA

- Each station multiplies the corresponding number by its chip (its orthogonal sequence), which is unique for each station.
- The result is a new sequence which is sent to the channel.
- For simplicity, we assume that all stations send the resulting sequences at the same time.
- The sequence on the channel is the sum of all four sequences as defined before.
- Now imagine station 3, which we said is silent, is listening to station 2.

# CDMA

- Station 3 multiplies the total data on the channel by the code for station 2, which is  $[+1 -1 +1 -1]$ , to get
- $[-1 -1 -3 +1] \cdot [+1 -1 +1 -1] = -4/4 = -1 \dots \dots \text{bit } 1$
- As Shown in figure.



# CDMA

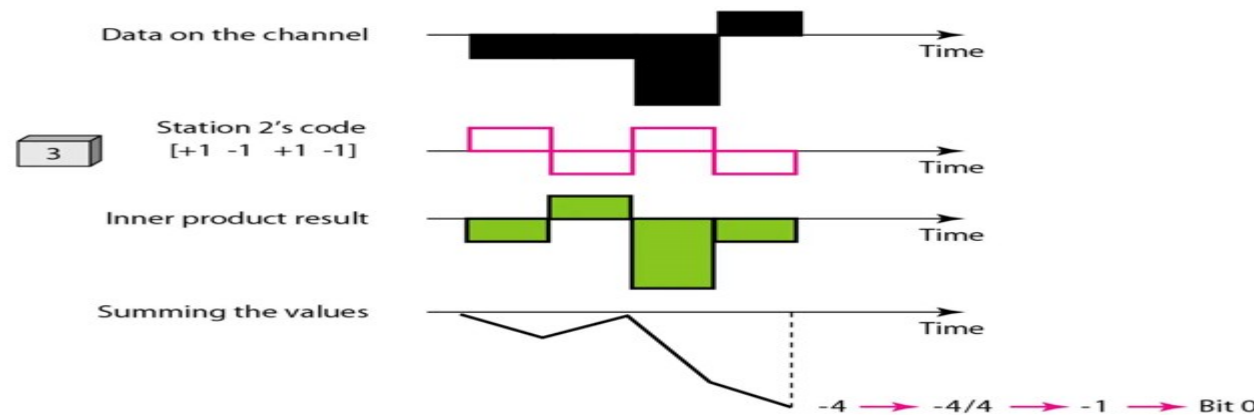
## Signal Level

- The process can be better understood if we show the digital signal produced by each station and the data recovered at the destination.
- The figure shows the corresponding signals for each station and the signal that is on the common channel.



# CDMA

- Figure shows how station 3 can detect the data sent by station 2 by using the code for station 2.
- The total data on the channel are multiplied (inner product operation) by the signal representing station 2 chip code to get a new signal.
- The station then integrates and adds the area under the signal, to get the value -4, which is divided by 4 and interpreted as bit 0.



# CDMA

## Sequence Generation

- To generate chip sequences, we use a Walsh table, which is a two-dimensional table with an equal number of rows and columns, as shown in Figure.
- The number of sequences in a Walsh table needs to be  $N = 2^m$ .

$$W_1 = [+1] \qquad W_{2N} = \begin{bmatrix} W_N & W_N \\ W_N & \overline{W_N} \end{bmatrix}$$

a. Two basic rules

$$W_1 = [+1]$$
$$W_2 = \begin{bmatrix} +1 & +1 \\ +1 & -1 \end{bmatrix}$$
$$W_4 = \begin{bmatrix} +1 & +1 & +1 & +1 \\ +1 & -1 & +1 & -1 \\ +1 & +1 & -1 & -1 \\ +1 & -1 & -1 & +1 \end{bmatrix}$$

b. Generation of  $W_1$ ,  $W_2$ , and  $W_4$

# CDMA

- Example 1
- Find the chips for a network with
  - a. Two stations
  - b. Four stations
- Solution
- We can use the rows of W2 and W4 in Figure
  - a. For a two-station network, we have  $[+1 \ +1]$  and  $[+1 \ -1]$ .
  - b. For a four-station network we have  $[+1 \ +1 \ +1 \ +1]$ ,  $[+1 \ -1 \ +1 \ -1]$ ,  $[+1 \ +1 \ -1 \ -1]$ , and  $[+1 \ -1 \ -1 \ +1]$ .

# CDMA

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